

[Soal] Geometry Everyday Project

AZZAM LABIB HAKIM |INSTAGRAM: @HAXUV_WORLD |THREADS: @AZZAM_29_12

March 22, 2026

Daftar Isi

1 Soal	3
1.1 Korea MO 2022, Problem 1	3
1.2 Japan MO 2022, Problem 3	3
1.3 IMO 2014, Problem 4	3
1.4 USA January TST for IMO 2016, Problem 3	3
1.5 Singapore MO Open 2022	3
1.6 Iran-Ukraine Camp 2021 Exam I P-1	4
1.7 USAMO 2015, Problem 2	4
1.8 All-Russian 2022 Grade 11 Problem 3	4
1.9 All-Russian 2022 Grade 9 Problem 2	4
1.10 CGMO 2022, Problem 3	5
1.11 IMSO 2021	5
1.12 Philippine MO 2022, Problem 6	5
1.13 Finnish National High School Competition 2019, Problem 3	5
1.14 Finnish National High School Competition 2001, Problem 1	5
1.15 Iberoamerican 2022, Problem 5	6
1.16 Random Problem by Wildan Bagus Wicaksono	6
1.17 USAJMO 2023, Problem 2	6
1.18 Korea 2023, Problem 1	6
1.19 IMO 2023, Problem 2	6
1.20 Japan 2023, Problem 2	7
1.21 Random Problem 2	7

1.22 LMNAS SMP 33 Penyisihan nomor 16	7
1.23 LMNAS SMP 33 Penyisihan nomor 3 isian singkat	7
1.24 LMNAS SMP 31 Semifinal nomor 17	7
1.25 Pelatda Pra OSP SMA 2024 DKI Jakarta	7
1.26 IMO 2024, Problem 4	8
1.27 Random Problem 3	8
1.28 KTOM Januari 2025, Nomor 3 Esai	8
1.29 OSN SMP 2016, Nomor 3, Hari Pertama	8
1.30 Random Problem 4	8
1.31 A lemma from Berkeley Math Tournament 2024	9
1.32 Japan MO 2025, Problem 2	9
1.33 OSP SMA 2019, Essay, Problem 5	9
1.34 British MO 2019 Round 2, Problem 1	9
1.35 JBMO 2018, Problem 4	9
1.36 Singapore MO Open 2025, Problem 1	10
1.37 Simulasi Semifinal OSN SMP 2025 DKI Jakarta, Uraian, Nomor 1	10
1.38 Kompetisi Sains Ruangguru 2021 Nomor 1	10
1.39 Iran MO 2018 Round 1, Problem 20	10
1.40 Random Problem 5, Classical Incenter Altitude	10
1.41 OSS (Seleksi Sekolah) Pradipta Dirgantara 2026	11

§1 Soal

§1.1 Korea MO 2022, Problem 1

Let ABC be an acute triangle with circumcenter O , and let D , E , and F be the feet of altitudes from A , B , and C to sides BC , CA , and AB , respectively. Denote by P the intersection of the tangents to the circumcircle of ABC at B and C . The line through P perpendicular to EF meets AD at Q , and let R be the foot of the perpendicular from A to EF . Prove that DR and OQ are parallel.

§1.2 Japan MO 2022, Problem 3

In an isosceles triangle ABC , $AB = AC$. Take a point O inside the triangle (not on the boundary). Draw a circle ω with center O and passing through C . Suppose $\omega \cap BC = \{C, D\}$, $\omega \cap AC = \{C, E\}$. Denote the circumcircle of $\triangle AEO$ as Γ . And suppose $\Gamma \cap \omega = \{E, F\}$. Prove that the circumcenter of $\triangle BDF$ is on Γ .

§1.3 IMO 2014, Problem 4

Let P and Q be on segment BC of an acute triangle ABC such that $\angle PAB = \angle BCA$ and $\angle CAQ = \angle ABC$. Let M and N be the points on AP and AQ , respectively, such that P is the midpoint of AM and Q is the midpoint of AN . Prove that the intersection of BM and CN is on the circumference of triangle ABC .

§1.4 USA January TST for IMO 2016, Problem 3

Let ABC be an acute scalene triangle and let P be a point in its interior. Let A_1 , B_1 , C_1 be projections of P onto triangle sides BC , CA , AB , respectively. Find the locus of points P such that AA_1 , BB_1 , CC_1 are concurrent and $\angle PAB + \angle PBC + \angle PCA = 90^\circ$.

§1.5 Singapore MO Open 2022

For $\triangle ABC$ and its circumcircle ω , draw the tangents at B, C to ω meeting at D . Let the line AD meet the circle with center D and radius DB at E inside $\triangle ABC$. Let F be the point on the extension of EB and G be the point on the segment EC such that $\angle AFB = \angle AGE = \angle A$. Prove that the tangent at A to the circumcircle of $\triangle AFG$ is parallel to BC .

§1.6 Iran-Ukraine Camp 2021 Exam I P-1

Incircle of the triangle $\triangle ABC$ is tangent to sides BC, AC, AB at points D, E, F respectively. Let I be the incentre of triangle and K be the intersection point of AI, BC . Circumcircle w of the triangle $\triangle IDK$ intersects IB and IC at points P and Q . Prove that lines EF, PQ and the tangent to w from D are concurrent.

§1.7 USAMO 2015, Problem 2

Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let X be a variable point on segment $\overline{PQ}$. Line AX meets ω again at S (other than A). Point T lies on arc AQB of ω such that $\overline{XT}$ is perpendicular to $\overline{AX}$. Let M denote the midpoint of chord $\overline{ST}$. As X varies on segment $\overline{PQ}$, show that M moves along a circle.

§1.8 All-Russian 2022 Grade 11 Problem 3

An acute-angled triangle ABC is fixed on a plane with largest side BC . Let PQ be an arbitrary diameter of its circumscribed circle, and the point P lies on the smaller arc AB , and the point Q is on the smaller arc AC . Points X, Y, Z are feet of perpendiculars dropped from point P to the line AB , from point Q to the line AC and from point A to line PQ . Prove that the center of the circumscribed circle of triangle XYZ lies on a fixed circle.

Terjemahan: Diberikan sebuah segitiga lancip ABC yang tetap di sebuah bidang dengan sisi terpanjangnya adalah BC . Misalkan PQ adalah sebuah diameter sembarang dari lingkaran luar segitiga ABC , dengan titik P berada di busur AB yang lebih kecil dan titik Q berada di busur AC yang lebih kecil. Titik X, Y, Z berturut-turut adalah kaki tinggi titik P ke garis AB , titik Q ke garis AC , dan titik A ke garis PQ . Buktikan bahwa pusat lingkaran luar segitiga XYZ berada pada sebuah lingkaran tetap.

§1.9 All-Russian 2022 Grade 9 Problem 2

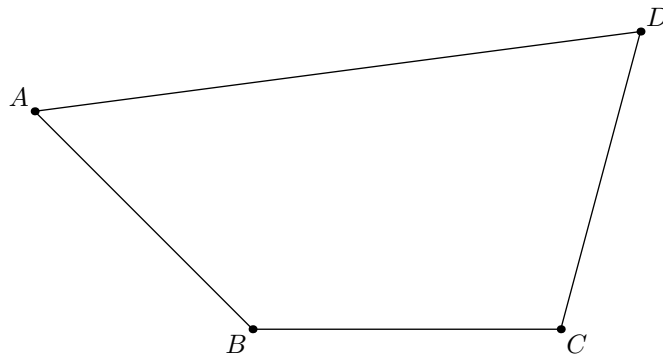
Given triangle ABC with incenter I and A -excenter J . Circle ω_b centered at point O_b passes through point B and is tangent to line CI at point I . Circle ω_c with center O_c passes through point C and touches line BI at point I . Let O_bO_c and IJ intersect at point K . Find the ratio IK/KJ .

§1.10 CGMO 2022, Problem 3

In triangle ABC , $AB > AC$, I is the incenter, AM is the midline. The line crosses I and is perpendicular to BC intersect AM at point L , and the symmetry of I with respect to point A is J Prove that: $\angle ABJ = \angle LBI$.

§1.11 IMSO 2021

In the diagram below, $AB = BC = CD$, $\angle ABC = 135^\circ$ and $\angle BCD = 105^\circ$. What is the measure, in degrees, of $\angle ADC$?

**§1.12 Philippine MO 2022, Problem 6**

In $\triangle ABC$, let D be the point on side BC such that $AB + BD = DC + CA$. The line AD intersects the circumcircle of $\triangle ABC$ again at point $X \neq A$. Prove that one of the common tangents of the circumcircles of $\triangle BD X$ and $\triangle CD X$ is parallel to BC .

§1.13 Finnish National High School Competition 2019, Problem 3

Let $ABCD$ be a cyclic quadrilateral whose side AB is at the same time the diameter of the circle. The lines AC and BD intersect at point E and the extensions of lines AD and BC intersect at point F . Segment EF intersects the circle at G and the extension of segment EF intersects AB at H . Show that if G is the midpoint of FH , then E is the midpoint of GH .

§1.14 Finnish National High School Competition 2001, Problem 1

In the right triangle ABC , CF is the altitude based on the hypotenuse AB . The circle centered at B and passing through F and the circle with centre A and the same radius intersect at a point of CB . Determine the ratio $FB : BC$.

§1.15 Iberoamerican 2022, Problem 5

Let ABC be an acute triangle with circumcircle Γ . Let P and Q be points in the half plane defined by BC containing A , such that BP and CQ are tangents to Γ and $PB = BC = CQ$. Let K and L be points on the external bisector of the angle $\angle CAB$, such that $BK = BA, CL = CA$. Let M be the intersection point of the lines PK and QL . Prove that $MK = ML$.

§1.16 Random Problem by Wildan Bagus Wicaksono

Diberikan persegi panjang $ABCD$. Titik P adalah perpotongan garis BC dengan garis yang melalui A dan tegak lurus AC . Titik Q pada segmen CD . Garis PQ memotong garis AD di R . Garis BR memotong garis AQ di X . Buktikan bahwa jika titik Q bergerak sepanjang segmen CD , maka besar $\angle BXC$ konstan.

(Authored by Wildan Bagus Wicaksono)

§1.17 USAJMO 2023, Problem 2

In an acute triangle ABC , let M be the midpoint of $\overline{BC}$. Let P be the foot of the perpendicular from C to AM . Suppose that the circumcircle of triangle ABP intersects line BC at two distinct points B and Q . Let N be the midpoint of $\overline{AQ}$. Prove that $NB = NC$.

§1.18 Korea 2023, Problem 1

In a triangle ABC ($\overline{AB} < \overline{AC}$), points $D(\neq A, B)$ and $E(\neq A, C)$ lies on side AB and AC respectively. Point P satisfies $\overline{PB} = \overline{PD}, \overline{PC} = \overline{PE}$. $X(\neq A, C)$ is on the arc AC of the circumcircle of triangle ABC not including B . Let $Y(\neq A)$ be the intersection of circumcircle of triangle ADE and line XA . Prove that $\overline{PX} = \overline{PY}$.

§1.19 IMO 2023, Problem 2

Let ABC be an acute-angled triangle with $AB < AC$. Let Ω be the circumcircle of ABC . Let S be the midpoint of the arc CB of Ω containing A . The perpendicular from A to BC meets BS at D and meets Ω again at $E \neq A$. The line through D parallel to BC meets line BE at L . Denote the circumcircle of triangle BDL by ω . Let ω meet Ω again at $P \neq B$. Prove that the line tangent to ω at P meets line BS on the internal angle bisector of $\angle BAC$.

§1.20 Japan 2023, Problem 2

In an acute triangle ABC , let D, E, F be the midpoints of BC, CA, AB respectively. Let X, Y be the feet of altitude from D to AB, AC respectively. The line through F parallel to XY meets line DY at P . Prove that $AD \perp EP$.

§1.21 Random Problem 2

Diberikan segitiga ABC dengan besar sudut $\angle ABC = 60^\circ$ dan $\angle BCA = 100^\circ$. Pada sisi AB dan AC berturut-turut terdapat titik D dan E sehingga $\angle EDC = 2\angle BCD = 2\angle CAB$. Tentukan besar sudut $\angle BED$.

§1.22 LMNAS SMP 33 Penyisihan nomor 16

Diberikan $\triangle ABC$ dengan $AB = AC$. Lingkaran dalam $\triangle ABC$ menyinggung sisi AB dan BC berturut-turut di P dan Q . Garis CP memotong lingkaran dalam $\triangle ABC$ di titik P dan R . Jika $PQ = 4$ dan $PR = 5$, panjang QR adalah ...

- a. $\frac{1}{2}\sqrt{14}$ b. 2 c. $\frac{3}{5}\sqrt{14}$ d. $\frac{4}{5}\sqrt{14}$ e. 3

§1.23 LMNAS SMP 33 Penyisihan nomor 3 isian singkat

Diberikan segitiga tumpul ABC dengan $AB = BC$. Titik D berada di dalam segitiga tersebut sedemikian sehingga $AD = BD$, $\angle ADB = 140^\circ$, dan $\angle ADC = 150^\circ$. Besar sudut ACD dalam satuan $^\circ$ (derajat) adalah ...

§1.24 LMNAS SMP 31 Semifinal nomor 17

Diberikan segitiga lancip ABC . Garis $\ell_1, \ell_2 \perp AB$ berturut-turut pada A dan B . Misalkan M titik tengah AB . Garis melalui M tegak lurus AC dan BC memotong ℓ_1 dan ℓ_2 berturut-turut pada E dan F . Misalkan D titik potong dari EF dan MC . Jika $EF = \frac{64}{3}$, $FM = 16$, dan $BD = 12$. Luas segitiga AMF dapat dinyatakan dalam $a\sqrt{b}$ dimana a dan b bilangan asli dan b tidak dapat dibagi oleh bilangan kuadrat sempurna selain 1. Nilai dari $a + b$ adalah ...

§1.25 Pelatda Pra OSP SMA 2024 DKI Jakarta

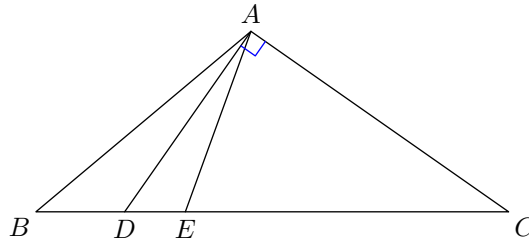
Pada segitiga ABC , titik D adalah kaki garis tinggi dari C . Titik E dan F pada AC dan BC , berturut-turut $AE = AD$ dan $BF = BD$. Titik S adalah refleksi C pada titik pusat lingkaran luar $\triangle ABC$. Buktikan bahwa $SE = SF$.

§1.26 IMO 2024, Problem 4

Let ABC be a triangle with $AB < AC < BC$. Let the incenter and incircle of triangle ABC be I and ω , respectively. Let X be the point on line BC different from C such that the line through X parallel to AC is tangent to ω . Similarly, let Y be the point on line BC different from B such that the line through Y parallel to AB is tangent to ω . Let AI intersect the circumcircle of triangle ABC at $P \neq A$. Let K and L be the midpoints of AC and AB , respectively. Prove that $\angle KIL + \angle YPX = 180^\circ$.
(Proposed by Dominik Burek, Poland)

§1.27 Random Problem 3

In the diagram below, AD is perpendicular to AC and $\angle BAD = \angle DAE = 15^\circ$. If $AB + AE = BC$, find $\angle ABC$ in degrees.

**§1.28 KTOM Januari 2025, Nomor 3 Esai**

Diberikan persegi $ABCD$ dan titik E pada ruas CD . Misalkan F adalah titik pada ruas AE sehingga BF tegak lurus AE . Dengan cara serupa, misalkan G adalah titik pada ruas EB sehingga AG tegak lurus EB . Misalkan garis DF dan CG berpotongan di H . Buktikan bahwa

$$\angle DFC + \angle DGC = 90^\circ + \angle AHD + \angle BHC.$$

§1.29 OSN SMP 2016, Nomor 3, Hari Pertama

Pada segitiga ABC , titik P dan Q berada pada sisi BC sehingga panjang BP sama dengan CQ . $\angle BAP = \angle CAQ$ dan $\angle APB$ lancip. Apakah segitiga ABC sama kaki? Tulis alasan Anda.

§1.30 Random Problem 4

Diberikan segitiga ABC dan lingkaran luar ω dari segitiga tersebut. Ditarik sembarang garis yang sejajar BC , yang memotong sisi AB dan AC secara berurutan di titik X dan Y . Misalkan M adalah titik tengah dari busur BC pada lingkaran ω yang tidak memuat titik A . Garis MX dan MY

memotong lingkaran ω kembali di titik S dan T secara berurutan. Misalkan ST dan XY berpotongan di titik P . Tunjukkan bahwa garis AP menyinggung lingkaran ω .

§1.31 A lemma from Berkeley Math Tournament 2024

The incircle of scalene triangle $\triangle ABC$ is tangent to $\overline{BC}$, $\overline{AC}$, and $\overline{AB}$ at points D , E , and F , respectively. The line EF intersects line BC at P and line AD at Q . The circumcircle of $\triangle AEF$ intersects line AP again at point $R \neq A$. If I is the incenter of $\triangle ABC$, prove that I, Q, R are collinear.

§1.32 Japan MO 2025, Problem 2

Let ABC be an acute triangle and let O be its circumcenter. Let O_1 and O_2 be the circumcenters of triangles ABO and ACO , respectively. The circumcircle of triangle AO_1O_2 intersects the side BC (excluding its endpoints) at two distinct points P and Q , such that the four points B, P, Q, C are collinear in this order. Let O_3 be the circumcenter of triangle OPQ . Prove that the three points A, O, O_3 are collinear.

§1.33 OSP SMA 2019, Essay, Problem 5

Diberikan segitiga ABC , dengan $AC > BC$, dan lingkaran luarnya yang berpusat di O . Misalkan M adalah titik pada lingkaran luar segitiga ABC sehingga CM adalah garis bagi $\angle ACB$. Misalkan Γ adalah lingkaran berdiameter CM . Garis bagi $\angle BOC$ dan garis bagi $\angle AOC$ memotong Γ berturut-turut di P dan Q . Jika K adalah titik tengah CM , buktikan bahwa P, Q, O, K terletak pada satu lingkaran.

§1.34 British MO 2019 Round 2, Problem 1

Let ABC be a triangle. Let L be the line through B perpendicular to AB . The perpendicular from A to BC meets L at the point D . The perpendicular bisector of BC meets L at the point P . Let E be the foot of the perpendicular from D to AC .

Prove that triangle BPE is isosceles.

§1.35 JBMO 2018, Problem 4

Diberikan $\triangle ABC$ dimana A', B', C' berturut-turut adalah pencerminan A, B, C terhadap BC, CA, AB . Perpotongan lingkaran luar $\triangle ABB'$ dan $\triangle ACC'$ adalah A_1 . Definisikan B_1 dan C_1 secara serupa.

Buktikan bahwa AA_1 , BB_1 , dan CC_1 konkuren (bertemu di satu titik).

§1.36 Singapore MO Open 2025, Problem 1

In the triangle ABC , $\angle B > 90^\circ$, the incircle touches the sides BC and CA at D and E , respectively. The lines ED and AB intersect at P . The incircle of the triangle AEP touches the sides PE and AP at D_1 and E_1 , respectively. The lines E_1D_1 and AE intersect at P_1 . Suppose P, C, E, B are concyclic. Prove that BE is parallel to PP_1 .

§1.37 Simulasi Semifinal OSN SMP 2025 DKI Jakarta, Uraian, Nomor 1

Diketahui $ABCD$ adalah segiempat talibusur dimana AB dan CD berpotongan di E . Titik K dan L terletak pada CD dan AB berturut-turut sehingga $\angle KAD = \angle KBC$ dan $\angle LDA = \angle LCB$. Buktikan bahwa $EK = EL$.

§1.38 Kompetisi Sains Ruangguru 2021 Nomor 1

Diberikan segitiga sama kaki ABC dengan $AB = AC$. Titik D dan E berturut-turut merupakan titik tengah AC dan BC , dan M titik tengah DE . Jika AM memotong BC pada F , buktikan bahwa DF menyinggung lingkaran luar ADM .

§1.39 Iran MO 2018 Round 1, Problem 20

In the convex and cyclic quadrilateral $ABCD$, we have $\angle B = 110^\circ$. The intersection of AD and BC is E and the intersection of AB and CD is F . If the perpendicular from E to AB intersects the perpendicular from F to BC on the circumcircle of the quadrilateral at point P , what is $\angle PDF$ in degrees?

Pada segiempat siklis $ABCD$, $\angle B = 110^\circ$. Perpotongan AD dan BC adalah E dan perpotongan AB dan CD adalah F . Jika garis tegak lurus dari E ke AB memotong garis tegak lurus dari F ke BC di P , dengan titik P terletak pada lingkaran luar segiempat tersebut, berapakah besar $\angle PDF$ dalam derajat?

§1.40 Random Problem 5, Classical Incenter Altitude

Diberikan segitiga ABC dengan $AB \neq AC$. Misalkan titik I adalah pusat lingkaran dalam dan r adalah panjang jari-jari lingkaran dalam segitiga ABC . Definisikan X sebagai titik tengah BC dan Y sebagai perpotongan XI dengan garis tinggi dari A . Buktikan bahwa panjang $AY = r$.

§1.41 OSS (Seleksi Sekolah) Pradipta Dirgantara 2026

Diberikan segitiga ABC dengan $\angle CAB = 75^\circ$ dan $ABC = 15^\circ$. Titik D pada segmen BC sehingga $BD = CA$ dan E pada sinar CA sehingga $CE = BC$. Misalkan F titik tengah AB dan M perpotongan CF dengan BE . Tunjukkan bahwa CM dan DM menyinggung lingkaran luar segitiga AEF .